C 5646 (Pages : 3) Name.....

Reg. No.....

SECOND SEMESTER B.Sc. DEGREE EXAMINATION, MAY 2016

(CUCBCSS–UG)

(Complementary Course)

STS 2C 02—PROBABILITY DISTRIBUTIONS

Time : Three Hours Maximum : 80 Marks

Section A

Answer **all** *questions in one word.*
Each question carries 1 *mark.*

Name the following :

1. The moments of a random variable X about E (X).
2. The distribution of X, where $\log_e X$ follows normal distribution.
3. The random variable with only two possible values 0 and 1 with respective probabilities q and p.

Fill up the blanks :

4. For the constants a and b and for any two random variables X and Y, Cov[X + a,Y + b] is ————————.
5. X is a Poisson random variable with mean 5. Then V(X) = ————————.
6. X and Y are two random variables with bivariate distribution function, $F_{xy}(x, y)$, then the value of $F_{xy}(-\infty, y)$ is ————————.
7. In the expansion of $M_x(t)$ the coefficient of $\frac{t^r}{r!}$ is ————————.

Write True *or* False :

8. Coefficient of correlation between two random variables X and Y is $\frac{\text{Cov}(X,Y)}{V(X)\,V(Y)}$.
9. The mean and standard deviation of a Poisson random variable are same.
10. For a sequence of random variables, variances of the random variables should exit to apply WLLN.

(10 × 1 = 10 marks)

Turn over

Section B

Answer **all** *questions in one sentence each.*
Each one carries 2 *marks.*

11. State Cauchy-Schwartz inequality.
12. Define conditional expectation.
13. Find the mode of a binomial distribution with parameters (10, 0.4)
14. Define the bivariate mgf of X and Y.
15. Find c, if $p(x,y) = c(2x + 3y)$, where $x = 0,1$ and $y = 1,2$ is a joint p.m.f.
16. Define standard normal distribution.
17. Define Pareto distribution.

$(7 \times 2 = 14$ marks)

Section C

Answer any **three** *questions.*
Each one carries 4 *marks.*

18. For two random variables X and Y, Show that E (X) = E [E (X / Y)].
19. $f(x, y) = e^{-x-y}$, $x > 0$, $y > 0$ be the joint p.d.f. of (X,Y). Find the bivariate m.g.f. of (X,Y).
20. Obtain the characteristic function of $X \sim B(n, p)$
21. Express Gamma distribution as sum of independent exponential random variables.
22. Define convergence in probability and convergence in distribution.

$(3 \times 4 = 12$ marks)

Section D

Answer any **four** *questions.*
Each one carries 6 *marks.*

23. For two random variables X and Y, the joint p.d.f. :

$f(x,y) = 2 - x - y;\ 0 \le x \le 1, 0 \le y \le 1$
$= 0$, otherwise.

Find Cov (X, Y).

24. X is a random variable with continuous distribution. Show that the distribution of $Y = F(x)$, where $F(x)$ is the distribution function of X is U [0, 1].

25. Prove that $\mu_{r+1} = r\lambda\mu_{r-1} + \lambda\frac{d}{d\lambda}\mu_r$, where $\mu_{r-1}, \mu_r, \mu_{r+1}$ are the central moments of Poisson distribution with parameter λ.

26. If $X \sim N(\mu, \sigma)$, prove that the odd order central moments are zero.

27. If $X \sim N(\mu, \sigma)$, show that $Y = X^2$ follow gamma distribution with parameters $\left(\frac{1}{2}, \frac{1}{2}\right)$.

28. State and prove Bernoulli's law of large numbers.

(4 × 6 = 24 marks)

Section E

Answer any **two** *questions.*
Each one carries 10 *marks.*

29. Let X and Y are two random variables with joint p.d.f. $f(x, y) = 2 ; 0 < x < y < 1$. Find
    (i) Correlation between X and Y. (ii) V (X/Y = y).

30. Under some limiting conditions such as $n \to \infty$ and $p \to 0$ such that $np = \lambda$ (finite) prove that binomial distribution, B (n,p) tends to Poisson P (λ).

31. $X \sim N(\mu, \sigma)$, prove that the $(2k)^{th}$ central moment $\mu_{2k} = \sigma^2 (2k-1)\mu_{2(k-1)}$ for $k = 1, 2, 3....$.

32. State and prove Tchebychev's inequality. For a random variable with p.d.f. $f(x) = e^{-x} ; x \geq 0$, use Tchebychev's inequality to find a lower bound to the probability $P(|X-1| < 2)$.

(2 × 10 = 20 marks)